BILL OF MATERIALS · PER PANEL BOX

RFID Panel Box — Parts List

Build qty × (pack price ÷ pieces per pack) · Prices: Amazon PO 1811 (Oct 12, 2026), Amazon listings, Alibaba checkouts
Updated Oct 17, 2026 · rev 5 · matches drawing RPB-001 rev 8 and the illustrated schematic

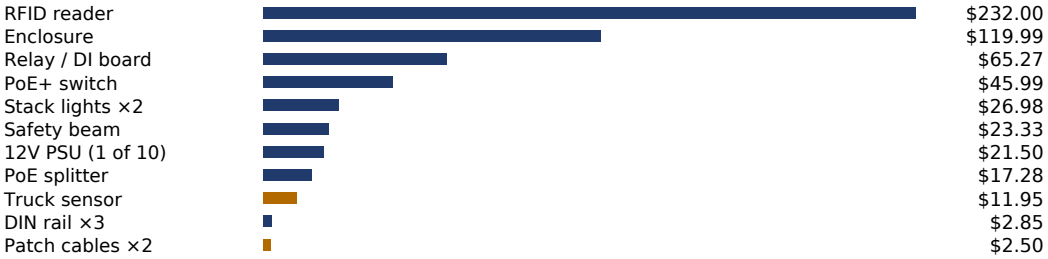
COST PER PANEL BOX	CONFIRMED PRICES	ESTIMATED PRICES	LONGEST LEAD TIME
$569.64	**$555.19**	**$14.45**	**Nov 13**
14 line items · 18 pieces · 2 unpriced (F1, C4) was $536.45 in rev 3 (+$33.19)	97% · PO 1811, Amazon, Alibaba	3% · DIHOOL sensor + patch cables	Alibaba reader + beam arrive Oct 23 – Nov 13

#	QTY	PART	VENDOR	PACK	PACK PRICE	UNIT	EXTENDED	PRICE FROM
1	1	**REVODATA PoE splitter (PS5712BG25)** 802.3bt 36 W, 12 V/3 A out, 5.5×2.5 plug · PO 1811, $17.28 ea	Amazon	1	$17.28	$17.28	**$17.28**	PO 1811
2	1	**NDR-120-12 DIN-rail power supply (PSU1)** 12 V 10 A 120 W · now feeds stack lights + sensors only · PO 1811 10-pack	Amazon	10	$215.00	$21.50	**$21.50**	PO 1811
3	1	**Waveshare ESP32-S3-POE-ETH-8DI-8RO (K1)** 8 relays + 8 isolated DI, PoE Ethernet (W5500) · replaces the 6-ch board · Amazon listing Oct 17	Amazon	1	$65.27	$65.27	**$65.27**	Amazon listing
4	3	**VAMRONE DIN rail, 12 in** 35 × 7.5 mm slotted aluminum · PO 1811 20-pack	Amazon	20	$18.99	$0.95	**$2.85**	PO 1811
5	2	**LED stack tower light with buzzer** 3-layer, 12–24 V · PO 1811 6-pack	Amazon	6	$80.95	$13.49	**$26.98**	PO 1811
6	1	**WYUAN M-808EK UHF RFID reader** 8 SMA ports · landed: $139.00 + $72.52 import + $13.74 tax + $6.74 fee	Alibaba	1	$232.00	$232.00	**$232.00**	Alibaba checkout
7	1	**JUTAI IRR-7M IR safety beam (B2)** Relay NO/NC, retro-reflective · landed: $15.00 + $5.89 import + $1.76 tax + $0.68 fee	Alibaba	1	$23.33	$23.33	**$23.33**	Alibaba checkout
8	1	**DIHOOL M16 laser photo sensor (B1)** Diffuse 3–80 cm, NPN NO, 10–30 VDC, 80 cm lead · Amazon ASIN B0F6X73HC3 · price seen at a reseller	Amazon	1	$11.95	$11.95	**$11.95**	Estimate
9	1	**GoodTop PoE+ booster switch, 5-port (SW1)** 802.3af/at, 4 × 30 W · includes 48 V DIN supply · Amazon listing	Amazon	1	$45.99	$45.99	**$45.99**	Amazon listing
10	2	**Rapink Cat6a patch cable, 0.5 ft** SW1→SPL1 and SW1→K1 (no spare) · 24-pack · not on PO 1811	Amazon	24	$29.99	$1.25	**$2.50**	Estimate
11	1	**TICONN IP67 enclosure, clear cover** 19.7 × 15.7 × 7.9 in, mounting plate · Amazon listing	Amazon	1	$119.99	$119.99	**$119.99**	Amazon listing
12	1	**48 V DIN power supply (PSU2)** Ships with the GoodTop switch · included in its $45.99	Amazon	1	$0.00	$0.00	**$0.00**	Included
13	1	**F1 — DIN fuse holder + fuse (AC line)** Feeds PSU1 and PSU2 · now required (was recommended) · not yet sourced	TBD	-	TBD	TBD	**TBD**	TBD
14	1	**C4 — Cat6 uplink cable, plant switch → G4 → SW1 P5** Length depends on the door · not yet sourced	TBD	-	TBD	TBD	**TBD**	TBD
		Total per panel box (priced items; F1 and C4 TBD)					**$569.64**	

Shaded rows changed since rev 3.

Where the money goes

■ Confirmed price ■ Estimated price



The RFID reader is 41% of the panel box cost.

Math. Unit = pack price ÷ pieces per pack; Extended = unit × qty. Amazon = PO 1811 or current listing, before tax/shipping (PO tax ≈6%). Alibaba = landed cost. **Estimates:** DIHOOL (reseller price) + patch cables = $14.45. **Added in rev 5:** F1 and C4, unpriced. **Dropped since rev 3:** ESP32-S3-Relay-6CH ($31.04) and MECCANIXITY M18 ($12.99 est.).

BEFORE YOU BUY

Compatibility Check

× 0 issue ! 9 to verify ✓ 8 OK

STATUS	AREA	FINDING	ACTION
! Check	**Network uplink**	Now designed: Cat6 from the plant switch through gland G4 to SW1 P5 (cable C4). Reader (TCP/IP) and K1 (MQTT) both depend on it. Length and price not set.	Run the cable to each door; add length and price to the BOM.
! Check	**Switch vs. PO 1811**	The panel uses the GoodTop PoE+ booster switch ($45.99) with its included 48 V DIN supply. PO 1811 bought 7× GoodTop PoE++ (802.3bt) at $38.61, which aren't in this design.	Order the PoE+ booster for each panel box; return or repurpose the PoE++ switches.
! Check	**PoE budget**	SW1 now powers two devices: P1 → splitter (reader) and P2 → K1. The Waveshare page doesn't give K1's PoE class or draw.	Confirm the switch's total PoE budget covers both ports; measure K1 on the bench.
! Check	**Reader power via splitter**	M-808EK is rated DC 9–24 V and asks for a 3 A supply (≈36 W at 12 V). On a 30 W PoE+ port the splitter gets ≤25.5 W, roughly 2 A at 12 V.	Test at full 33 dBm with all antennas. If it browns out, power the reader from PSU1 and use the splitter for data only.
! Check	**DC plug size**	Splitter output is a 5.5 × 2.5 mm plug. Many reader boards use 5.5 × 2.1 mm jacks.	Check the M-808EK jack; use a 2.5→2.1 mm adapter or land the leads on its power terminal.
! Check	**Sensor input wiring**	DIHOOL is NPN NO (BK → K1 DI1); the JUTAI beam is a dry contact (NO → DI2, COM → DGND). The drawing bonds K1 DGND to 0 V so PSU1-fed sensors share a reference. PO 1811's Heschen E3F-DS100P3 sensors are PNP.	Verify against Waveshare's DI wiring diagram on the bench. Don't swap in the PNP Heschen without re-wiring.
! Check	**Beam model**	The Alibaba checkout says IRR-7M; the product photos you have show IRR-7R. The drawing uses the ordered part and the wire colors from the wiring sheet.	Confirm the model and wire colors when it arrives.
! Check	**Truck sensor lead**	The DIHOOL has an 80 cm (≈2.6 ft) lead and mounts outside on the dock face near a bumper.	Plan a splice / junction box and extension cable to reach gland G3.
! Check	**Non-DIN parts**	The PoE splitter and the reader plate don't clip onto DIN rail.	Screw or bracket them to the backplate (PO 1811 includes C45 DIN PCB clips).
✓ OK	**Relay count**	K1 has 8 relays. Both towers in parallel use CH1–CH4 (red, amber, green, buzzer).	CH5–CH8 spare — or drive tower 2 on its own.
✓ OK	**Switch power**	SW1 runs from PSU2 (48 V, ships with SW1), inside its 12–52 V input range.	Shown as PSU2 on rail 1 in rev 7; F1 feeds PSU1 and PSU2.
✓ OK	**12 V power budget**	NDR-120-12 = 120 W. It now feeds only 2 stack lights (~10 W) and sensors (<1 W): about 11 W. K1 is PoE-powered.	Plenty of headroom to power the reader directly if needed.
✓ OK	**Relay / DI board**	ESP32-S3-POE-ETH-8DI-8RO: PoE + W5500 Ethernet, 8 relays (10 A at 250 VAC / 30 VDC), 8 isolated inputs, 175 × 90 mm.	Fits rail 2 beside TB1; verify before cutting rails.
✓ OK	**Reader GPIO**	Sensors moved to K1, so the M-808EK's 2 opto inputs + 2 opto outputs are free.	Keep for future use.
✓ OK	**Safety beam**	JUTAI IRR-7M: 10–30 VDC (12 V ✓), relay NO/NC rated 200 mA, ≤11 mA draw. Retro-reflective, so it needs a reflector.	Confirm the reflector ships with it.
✓ OK	**Enclosure & rails**	Rail 1: PSU1, PSU2, SW1, with SPL1 screwed to the plate beside them. Rail 2: K1 + TB1. 7.9 in depth clears the NDR-120.	3rd rail spare for fusing.
✓ OK	**Patch cables**	Two 0.5 ft cables: SW1 → splitter and SW1 → K1. The splitter's RJ45 pigtail goes to the reader.	No spare left in this count.

Not on the build list (needed to finish a panel box)

- UHF antennas + RF cable — none on the list; PO 1811 has BNC panel mounts and BNC→SMA RG316 pigtails
- AC power cord / supply drop to G1 — not on the list (F1 is now a BOM line)
- DIN terminal blocks — **TB1 is now required** (12 V distribution); PO 1811: GUETNEU terminal block kit
- Cable glands — PO 1811: QILIPSU ½" NPT glands (G1 AC, G2 beam, G3 truck sensor, G4 uplink, G5 stack lights)
- Stack-light cable — PO 1811: 18 AWG 5-conductor, 100 ft
- Truck-sensor extension cable + small junction box — not on the list

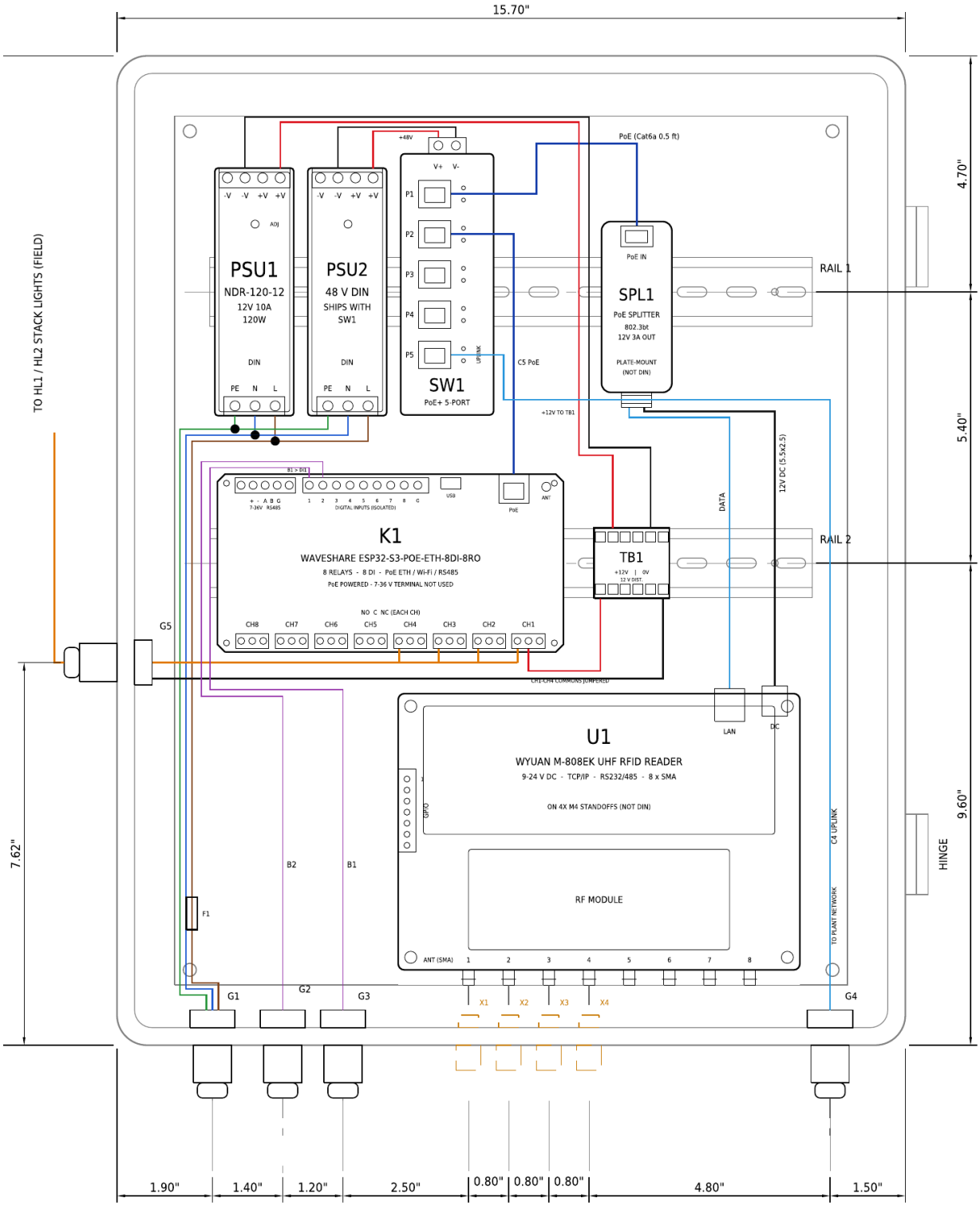
Return / repurpose from PO 1811: Waveshare ESP32-S3-Relay-6CH 2-pack · 7× GoodTop PoE++ switches.

Specs from product pages: M-808EK DC 9–24 V 3 A, 2 opto in / 2 opto out; JUTAI IRR-7M 10–30 VDC, relay NO/NC 200 mA; ESP32-S3-POE-ETH-8DI-8RO 8 relays ≤10 A, 8 DI, PoE port; DIHOOL NPN NO, 10–30 VDC, 3–80 cm.

REV 8 · MATCHES THE BUILD LIST

Panel Layout

Front view from drawing RPB-001 rev 8, sheet 1. PSU1 12 V: stack lights + sensors via TB1 · PSU2 48 V: SW1 · SW1 PoE: splitter (reader) on P1, K1 on P2 · sensors to K1 DI1 / DI2 · C4 uplink via G4 · F1 on the AC line.



FRONT VIEW - COVER REMOVED

Full sheets (layout + wiring schematic): RFID_Panel_Box_RPB-001_rev8.pdf / .dxf · illustrated schematic: RFID_Panel_Box_Illustrated_Schematic_rev8.